

A hybrid SOFM-SVR with a filter-based feature selection for stock market forecasting

Science Direct

Expert Systems with Applications xxx (2008) xxx–xxx

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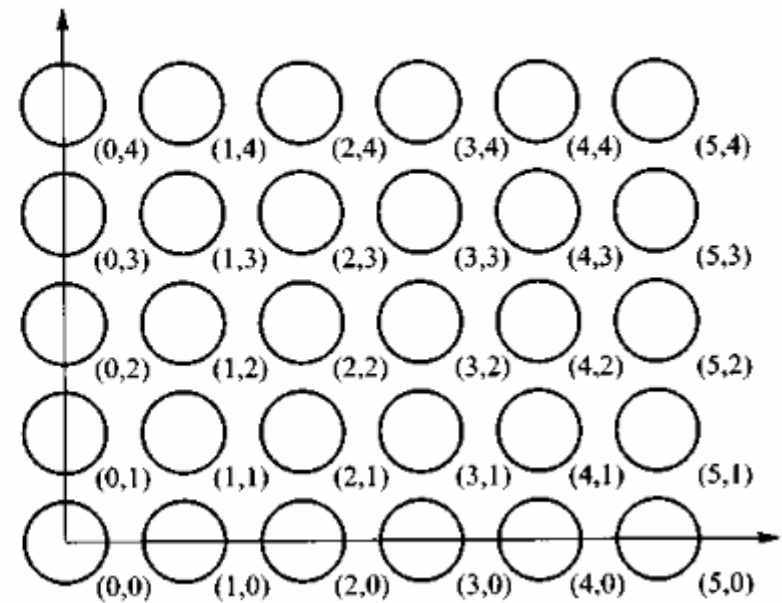
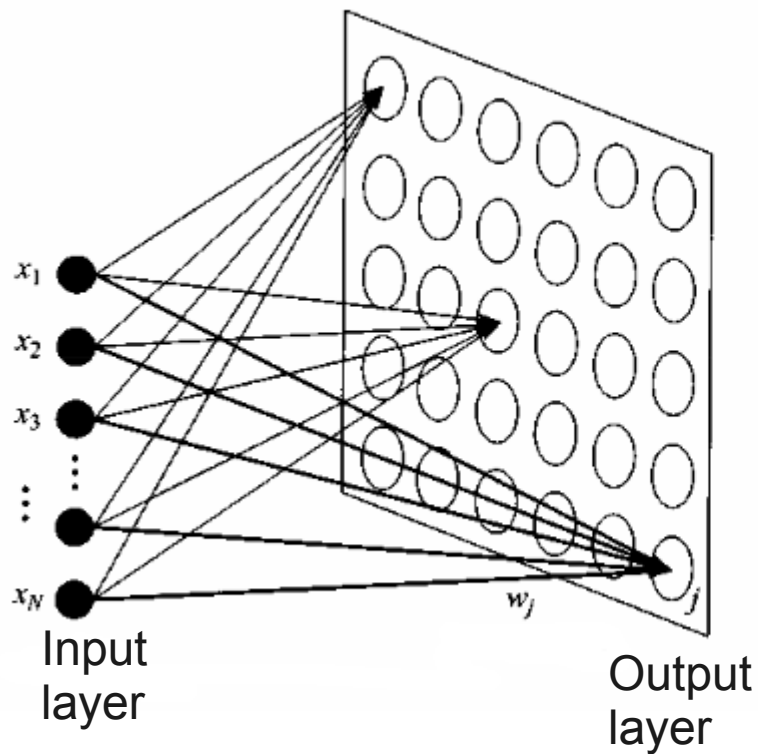
Date: Aug. 27, 2008

SOFM Example (1/13)

- SOM (or SOFM) is a type of artificial neural network that is trained using unsupervised learning to produce a low-dimensional (typically two dimensional), discretized representation of the input space of the training samples, called a map.
- The map (usually use Kohonen map, 1982) seeks to preserve the topological properties of the input space.

SOFM Example (2/13)

- Input layer: $X = [x_1, x_2, \dots, x_N]^T$
- Output layer: $W_j = [w_{j1}, w_{j2}, \dots, w_{jN}]^T$



SOFM Example (3/13)

- Center of the neighborhood

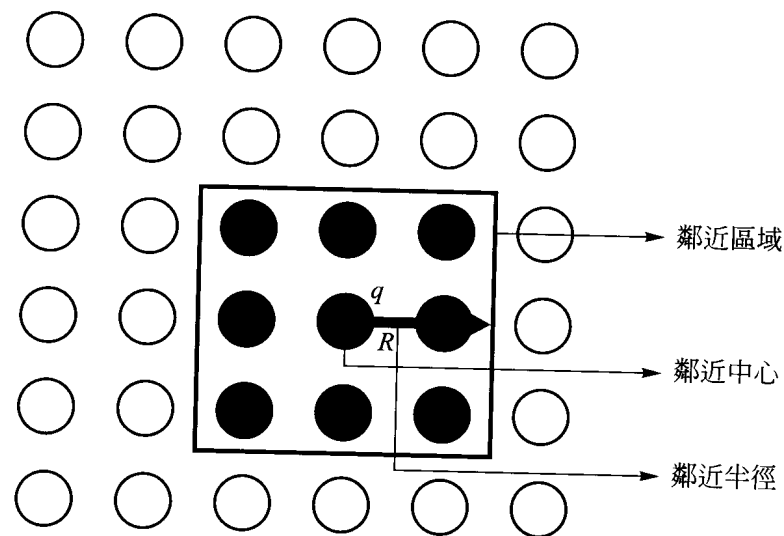
- winner node: $q(X) = \min_{\forall j} \|X - w_j\| = \sqrt{(X - w_{j1})^2 + \dots + (X - w_{jN})^2}$

- Neighborhood function

- $\eta_{qj} = \exp\left(\frac{-\|r_j - r_q\|^2}{R^2}\right)$ q : the winner R : neighbor radius

- Learning rule

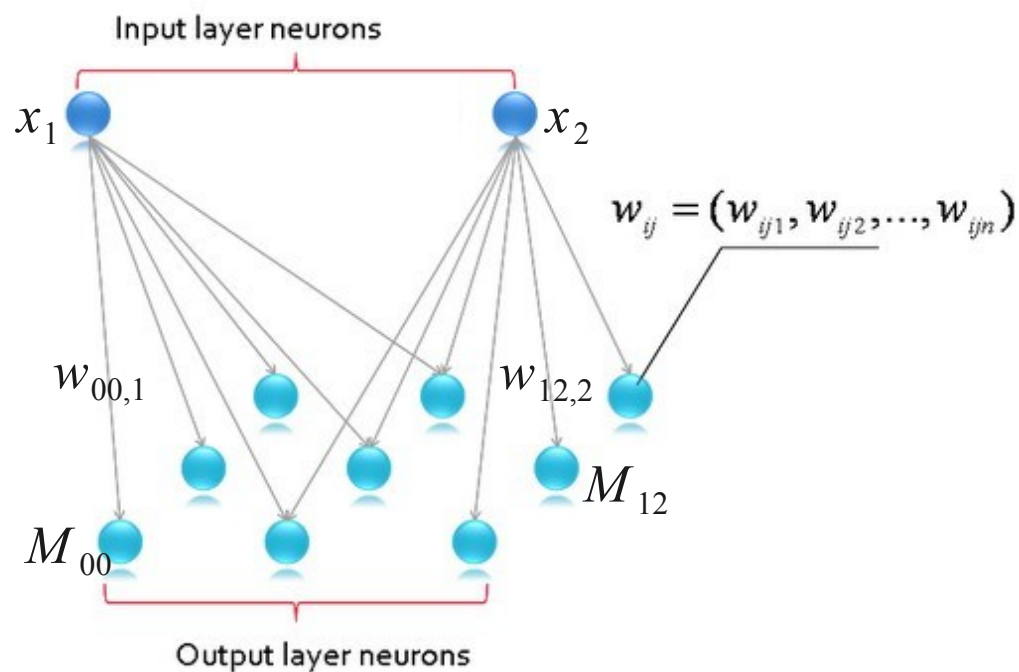
- $w_j(k) = w_j(k-1) + \Delta w_j(k)$
 - $\Delta w_j(k) = \mu(k) \eta(k) [x(k) - w_j(k)]$
 - $\mu(k)$: the learning rate function at k



SOFM Example (4/13)

- Environment setting: $M_{3 \times 3}$, $R=1$, $\mu=0.5$
- Input data: $x_i = (Gender_i, Height_i)$

X	Gender	Height
1	1(Female)	1.6
2	0(Male)	2
3	1	1.9
4	1	1.88
5	1	1.7
6	0	1.85
7	1	1.6
8	0	1.7
9	0	2.2
10	0	2.1
11	1	1.8
12	0	1.95
13	1	1.9
14	1	1.8
15	1	1.75



SOFM Example (5/13)

- Randomize the nodes' weight vector

$w(00,0)=0.71$ $w(01,0)=0.26$ $w(02,0)=0.56$

$w(10,0)=0.77$ $w(11,0)=0.16$ $w(12,0)=0.73$

$w(20,0)=0.19$ $w(21,0)=0.94$ $w(22,0)=0.95$

$w(00,1)=0.4$ $w(01,1)=0.44$ $w(02,1)=0.66$

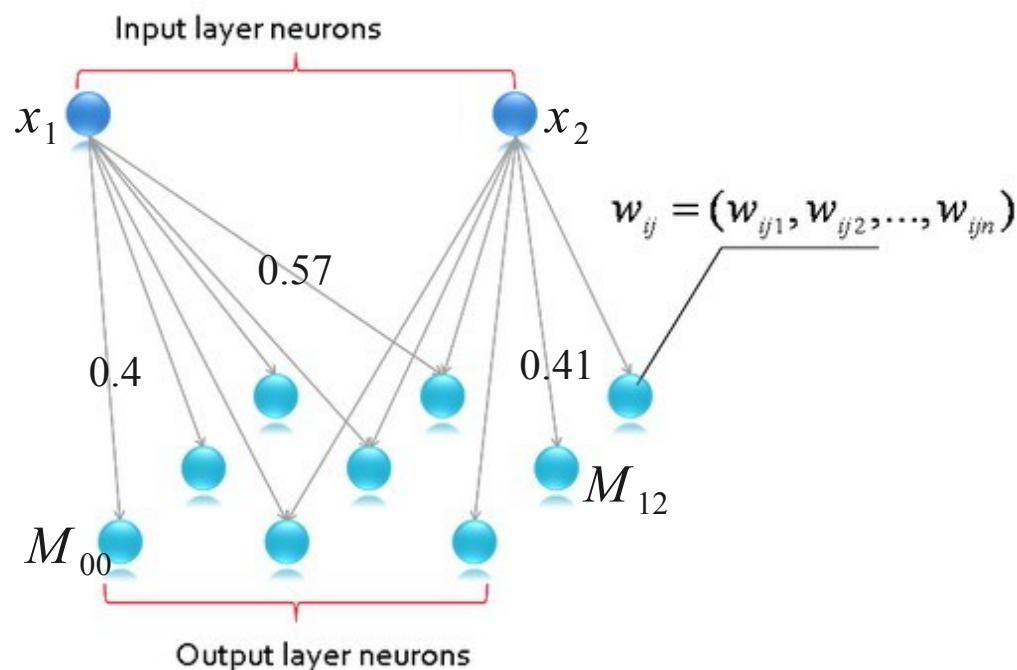
$w(10,1)=0.3$ $w(11,1)=0.06$ $w(12,1)=0.57$

$w(20,1)=0.39$ $w(21,1)=0.57$ $w(22,1)=0.85$

$w(00,2)=0.88$ $w(01,2)=0.06$ $w(02,2)=0.14$

$w(10,2)=0.84$ $w(11,2)=0.93$ $w(12,2)=0.41$

$w(20,2)=0.24$ $w(21,2)=0.64$ $w(22,2)=0.18$



SOFM Example (6/13)

- Input: $x_1 = (1, 1.6)$

- Similarity:

sim(00)=0.9625

sim(01)=3.2328

sim(02)=2.4408

sim(10)=1.1205

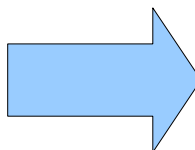
sim(11)=2.0381

sim(12)=1.6739

sim(20)=2.8778

sim(21)=1.1101

sim(22)=2.0414



	W(k-1)	$\Delta W(k)$	W(k)
00,0	0.71	0.15	0.86
00,1	0.4	0.3	0.7
00,2	0.88	0.36	1.24
01,0	0.26	0.37	0.63
01,1	0.44	0.28	0.72
01,2	0.06	0.77	0.83
10,0	0.77	0.12	0.89
10,1	0.3	0.35	0.65
10,2	0.84	0.38	1.22
11,0	0.16	0.15	0.31
11,1	0.06	0.17	0.23
11,2	0.93	0.12	1.05

- Winner: $q(X) = (0,0)$

SOFM Example (7/13)

- Input: $x_2=(0,2)$

- Similarity:

sim(00)=1.08863

sim(01)=2.0242

sim(02)=4.0888

sim(10)=1.04412

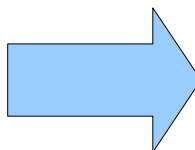
sim(11)=1.4205

sim(12)=2.9259

sim(20)=3.9058

sim(21)=2.1781

sim(22)=4.0374



- Winner: $q(X)=(1,0)$

	$W(k-1)$	$\Delta W(k)$	$W(k)$
10,0	0.89	0.06	0.94
10,1	0.65	-0.33	0.33
10,2	1.22	0.39	1.61
11,0	0.31	0.34	0.66
11,1	0.23	-0.12	0.12
11,2	1.05	0.47	1.53
20,0	0.19	0.41	0.6
20,1	0.39	-0.2	0.2
20,2	0.24	0.88	1.12
00,0	0.86	0.07	0.93
00,1	0.7	-0.35	0.35
00,2	1.24	0.38	1.62
21,0	0.94	0.01	0.95
21,1	0.57	-0.1	0.47
21,2	0.64	0.25	0.89
01,0	0.63	0.07	0.7
01,1	0.72	-0.13	0.59
01,2	0.83	0.22	1.05

SOFM Example (8/13)

- Input: $x_3 = (1, 1.9)$

- Similarity:

sim(00)=0.506156

sim(01)=0.99194

sim(02)=3.4068

sim(10)=0.543031

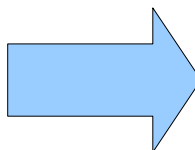
sim(11)=1.03754

sim(12)=2.4779

sim(20)=1.42045

sim(21)=1.30824

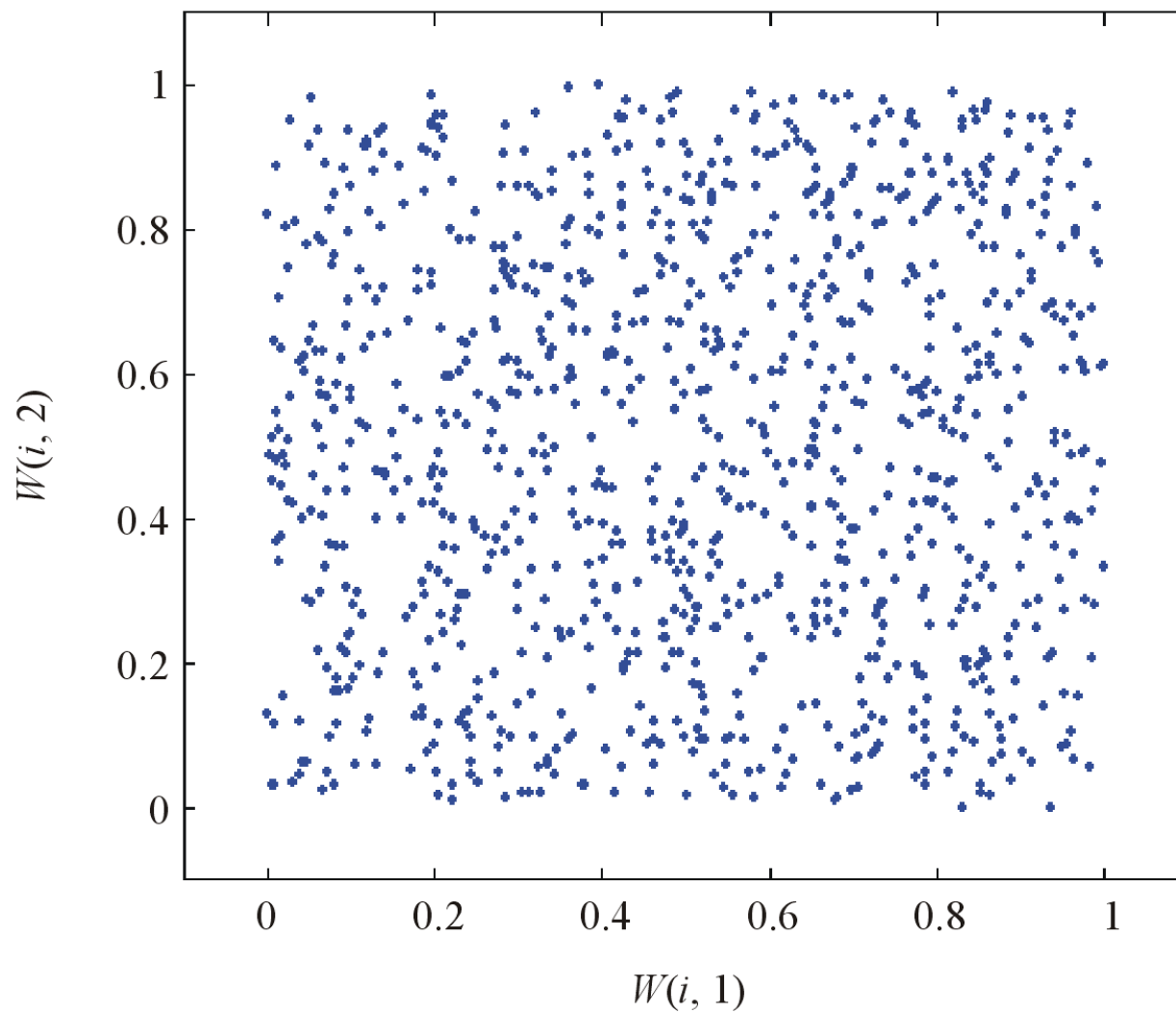
sim(22)=2.9834



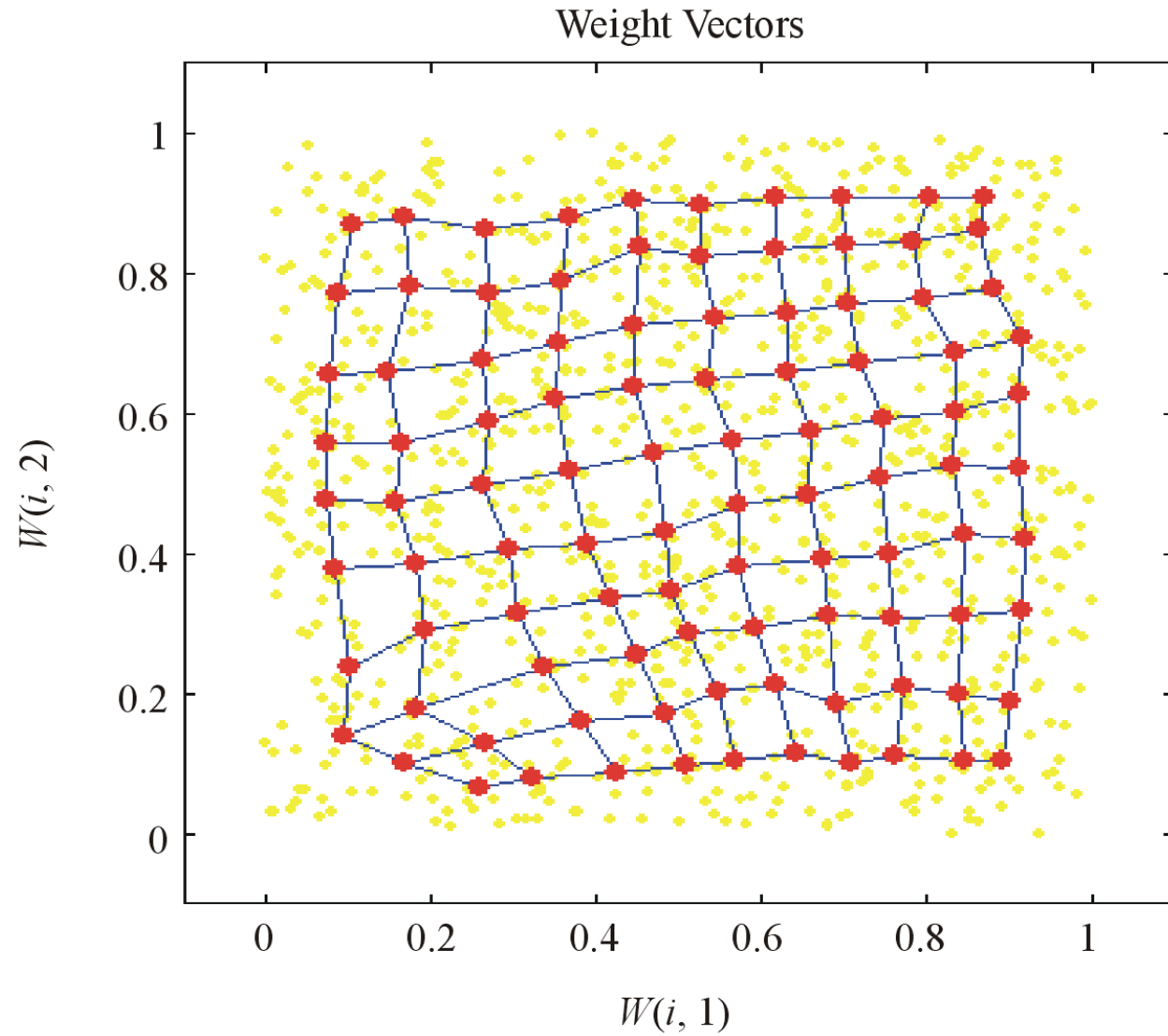
	W(k-1)	$\Delta W(k)$	W(k)
00,0	0.93	0.04	0.96
00,1	0.35	0.33	0.68
00,2	1.62	0.14	1.76
01,0	0.7	0.15	0.85
01,1	0.59	0.21	0.79
01,2	1.05	0.43	1.47
10,0	0.94	0.03	0.97
10,1	0.33	0.34	0.66
10,2	1.61	0.15	1.76
11,0	0.66	0.06	0.72
11,1	0.12	0.16	0.28
11,2	1.53	0.07	1.6

- Winner: $q(X) = (0, 0)$

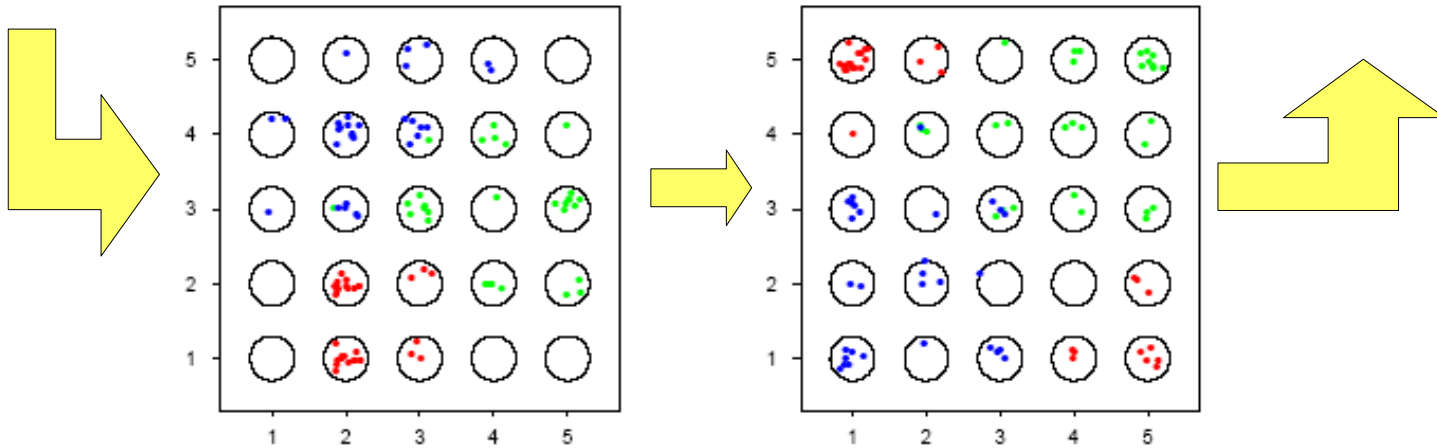
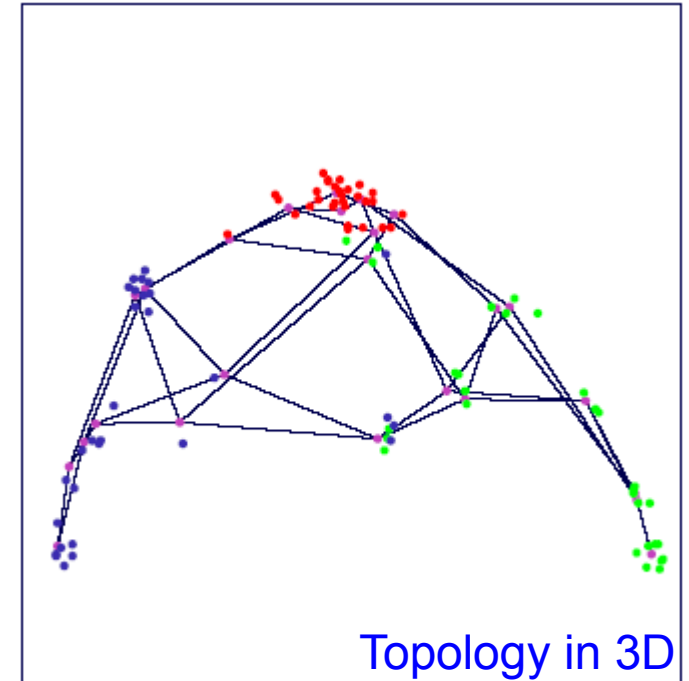
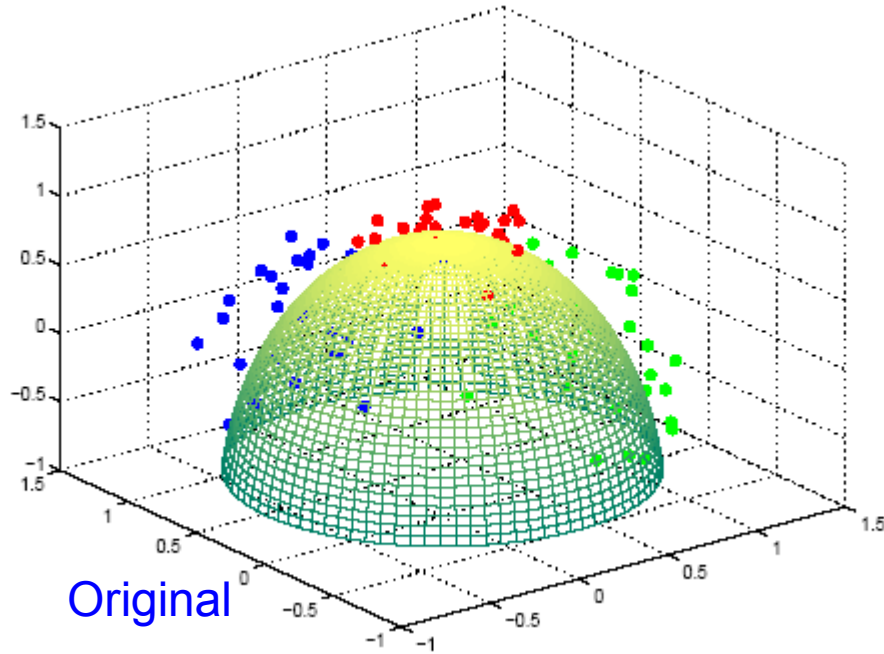
SOFM Example (9/13)



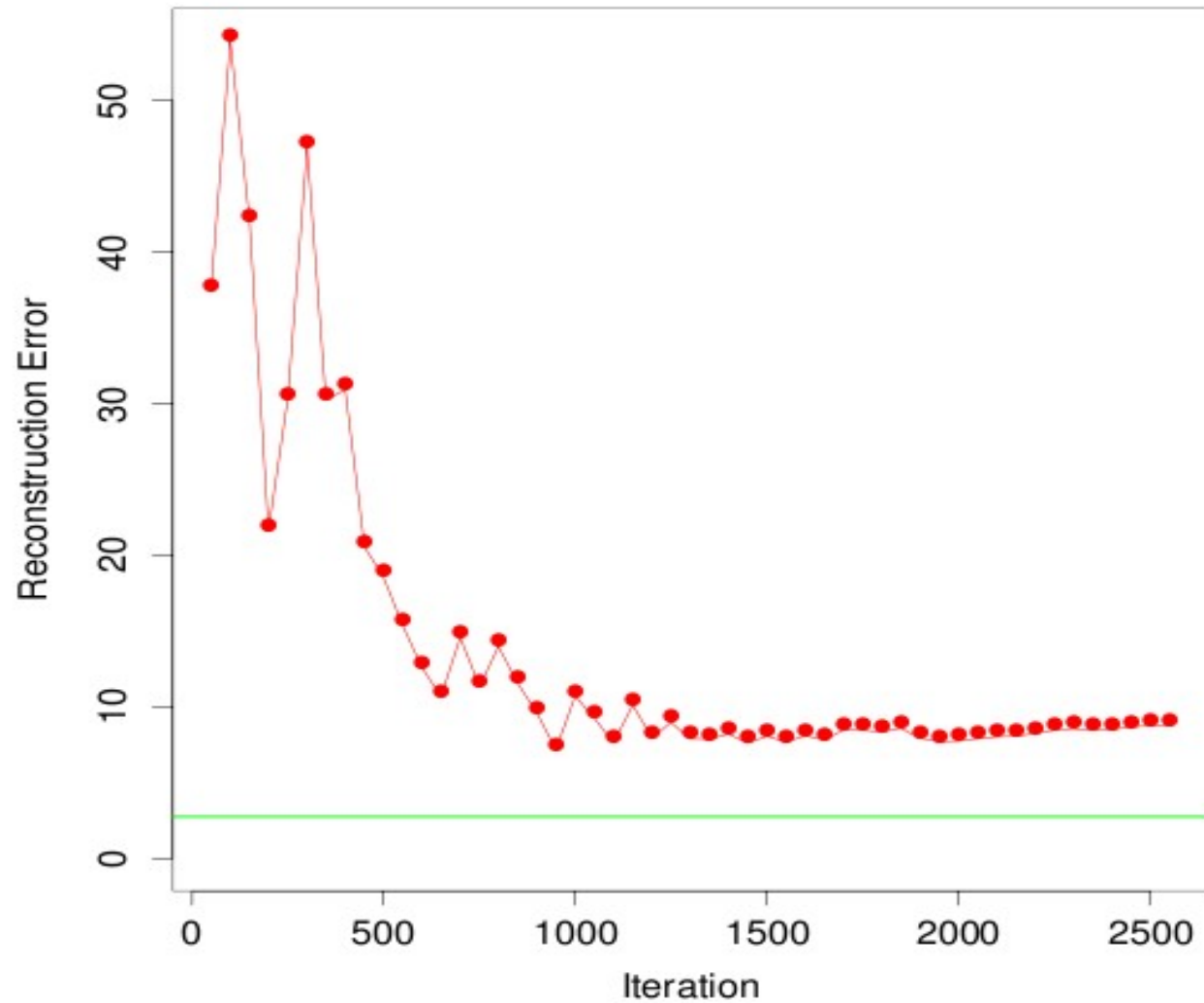
SOFM Example (10/13)



SOFM Example (11/13)



SOFM Example (12/13)



SOFM Example (13/13)

- Demo
 - <http://www.eas.asu.edu/~eee511/demos/som-color/s>
 - <http://www.patol.com/java/fill/index.html>
 - <http://staff.aist.go.jp/utsugi-a/Lab/BSOM1/index.html>